Markov's Inequality Scribe

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Generate a descriptive lecture scribe for exam preparation using the provided context. For every topic, reason through the definitions, step by step proofs, and examples to make the logical flow explicit for a student. Do not add outside material. Consider slides from 18 i.e. markov's inequality

Here is a **descriptive lecture scribe (exam-focused)** based strictly on the provided slides, focusing on **Markov's Inequality (from slide 18 onward)** and following the structure indicated in the outline .

Lecture Scribe: Markov's Inequality

1. Motivation: Tail Bounds

The lecture begins by motivating the need for **tail bounds**.

- In many computational systems (e.g., servers), we are not only interested in the *average behavior* but in **extreme cases**.
- Specifically, we want to understand:

  > *What is the probability that a random variable takes a very large value?*

This leads to the concept of bounding:

$$P(X \geq a)$$

for large a, which lies in the **tail** of the distribution.

Thus, **tail inequalities** provide guarantees on how unlikely extreme events are.

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2. Markov's Inequality – Definition

Markov's Inequality is the **first and simplest tail bound** introduced.

Definition

Let X be a **non-negative random variable** (i.e., $X \geq 0$).

Then for any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

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3. Understanding the Definition

Let us carefully reason through each component:

- The condition $X \geq 0$ is **essential**:
  - The inequality relies on the fact that values cannot cancel out (no negative values).
- The inequality relates:
  - A **probability of a large deviation** $P(X \geq a)$
  - To the **expected value** $\mathbb{E}[X]$

Key Insight

If the expectation is small, then the probability that X is very large must also be small.

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4. Proof of Markov's Inequality (Step-by-Step)

We now derive the result logically.

Step 1: Start with expectation definition

$$\mathbb{E}[X] = \sum_{x} x \cdot P(X = x)$$

Step 2: Focus only on large values

Split the sum into two parts:

- Values where $X < a$
- Values where $X \geq a$

$$\mathbb{E}[X] = \sum_{x<a} xP(X = x) + \sum_{x \geq a} xP(X = x)$$

Step 3: Lower bound the second term

For all $x \geq a$, we know:

$$x \geq a$$

So,

$$xP(X = x) \geq a \cdot P(X = x)$$

Thus,

$$\sum_{x \ge a} xP(X = x) \ge \sum_{x \ge a} a \cdot P(X = x)$$

$$= a \cdot \sum_{x \ge a} P(X = x)$$

$$= a \cdot P(X \ge a)$$

Step 4: Combine with expectation

Since expectation includes this term:

$$\mathbb{E}[X] \ge a \cdot P(X \ge a)$$

Step 5: Rearranging

$$P(X \ge a) \le \frac{\mathbb{E}[X]}{a}$$

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5. Key Observations

- The proof does **not depend on distribution type**.
- Only requires:
  - Non-negativity of X
  - Definition of expectation

Interpretation

Markov's inequality gives a **worst-case bound**:

- It may not be tight.
- But it is always valid.

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6. Example (As per Lecture Flow)

Consider a random variable X with:

$$\mathbb{E}[X] = 10$$

We want to bound:

$$P(X \ge 50)$$

Apply Markov's Inequality

$$P(X \ge 50) \le \frac{10}{50} = 0.2$$

Interpretation

- Even without knowing the distribution,
- We can guarantee:

  > The probability that X is at least 50 is **at most 0.2**.

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7. Intuition for Exams

- Think of expectation as "average mass".
- If too much probability were in large values,
  - The expectation would increase.
- Therefore, **large deviations must be rare**.

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8. Role in Tail Bounds Framework

Markov's Inequality is:

- The **foundation** for stronger inequalities:
  - Chebyshev's Inequality
  - Chernoff Bounds
- A **starting point** for bounding:
  - Worst-case system loads
  - Rare events in computing systems

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9. Connection to Case Study

From the lecture context:

- When predicting **worst-case server demand**:
  - Exact distribution may not be known
  - But expectation is often known

Markov's inequality allows:

$$\text{Bounding extreme demand probability using only } \mathbb{E}[X]$$

Final Summary

- Markov's Inequality bounds tail probability using expectation.
- Requires only:
  - $X \geq 0$

- Core result:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

- Proof relies on:
  - Splitting expectation
  - Lower bounding large-value contributions
- Provides:
  - A **simple but powerful worst-case guarantee**

If you want, I can continue the same **step-by-step scribe style** for:

- Chebyshev's Inequality
- Chernoff Bounds
- Jensen's Inequality

just like this.

yes everything from the markov inequality slides

Here is the **complete descriptive lecture scribe (exam-focused)** covering **all topics starting from Markov's Inequality slides onward**, following the exact logical flow of the lecture and strictly adhering to the provided content .

Lecture Scribe: Inequalities and Tail Bounds (From Markov's Inequality Onward)

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1. Markov's Inequality

1.1 Definition

Let X be a **non-negative random variable** ($X \geq 0$).

For any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

1.2 Step-by-Step Proof

Step 1: Start from expectation

$$\mathbb{E}[X] = \sum_{x} x \cdot P(X = x)$$

Step 2: Split based on threshold a

$$\mathbb{E}[X] = \sum_{x<a} xP(X = x) + \sum_{x \geq a} xP(X = x)$$

Step 3: Bound the large-value part

For all $x \geq a$,

$$x \geq a \Rightarrow xP(X = x) \geq aP(X = x)$$

Thus,

$$\sum_{x \geq a} xP(X = x) \geq a \sum_{x \geq a} P(X = x)$$

$$= a \cdot P(X \geq a)$$

Step 4: Combine

$$\mathbb{E}[X] \geq a \cdot P(X \geq a)$$

Step 5: Rearranging

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

1.3 Example

If:

$$\mathbb{E}[X] = 10$$

Then:

$$P(X \geq 50) \leq \frac{10}{50} = 0.2$$

1.4 Key Insight

- No assumption on distribution.
- Only requires non-negativity.
- Provides a **loose but universal bound**.

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2. Chebyshev's Inequality

2.1 Motivation

Markov's inequality only uses **expectation**.

To get a tighter bound, we incorporate **variance**.

2.2 Definition

For any random variable X with mean μ and variance σ^2,

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

2.3 Proof Using Markov's Inequality

Step 1: Define new variable

Let:

$$Y = (X - \mu)^2$$

- $Y \geq 0$, so Markov applies.

Step 2: Apply Markov

$$P(Y \geq a^2) \leq \frac{\mathbb{E}[Y]}{a^2}$$

Step 3: Substitute back

$$P((X-\mu)^2 \geq a^2) = P(|X-\mu| \geq a)$$

Step 4: Use variance definition

$$\mathbb{E}[Y] = \mathbb{E}[(X-\mu)^2] = \sigma^2$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

2.4 Example

If:

- $\mu = 100$
- $\sigma^2 = 25$

Then:

$$P(|X-100| \geq 10) \leq \frac{25}{100} = 0.25$$

2.5 Insight

- Uses both **mean and spread**.
- Stronger than Markov.
- Still distribution-independent.

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3. Chernoff Bound (Poisson Tail)

3.1 Motivation

- Markov and Chebyshev give **loose bounds**.
- Need **exponentially decreasing bounds** for sums of random variables.

3.2 Idea

Instead of applying Markov to X, apply it to:

$$e^{tX}$$

This amplifies large deviations.

3.3 Markov on Exponential

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

3.4 Resulting Bound

$$P(X \geq a) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

- Choose t to **minimize the bound**.

3.5 Interpretation

- Uses **moment generating function (MGF)**.
- Produces **exponentially tight bounds**.
- Especially useful for:
  - Sum of independent random variables
  - Poisson-type tails

3.6 Key Insight

- Transforming the variable makes rare events more visible.
- Leads to much stronger guarantees than Chebyshev.

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4. Jensen's Inequality

4.1 Convex Functions

A function f is **convex** if:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

for $0 \leq \lambda \leq 1$.

4.2 Definition of Jensen's Inequality

For a convex function f:

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

4.3 Proof Idea (Step-by-Step Reasoning)

Step 1: Use convexity

Convexity says function lies below line segments.

Step 2: Represent expectation as weighted sum

$$\mathbb{E}[X] = \sum p_i x_i$$

Step 3: Apply convexity repeatedly

$$f\left(\sum p_i x_i\right) \leq \sum p_i f(x_i)$$

Step 4: Recognize expectation

$$\sum p_i f(x_i) = \mathbb{E}[f(X)]$$

Final Result

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

4.4 Example Insight

- If $f(x) = x^2$ (convex):

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$$

4.5 Interpretation

- Applying function **after expectation** gives smaller value than:
- Taking expectation **after applying function**

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5. Case Study: Worst-Case Server Demand

5.1 Problem Context

- Predict **maximum load** on a server.
- Exact distribution unknown.

5.2 Approach Using Inequalities

Step 1: Use expectation

- Apply **Markov** for rough bound.

Step 2: Use variance

- Apply **Chebyshev** for tighter bound.

Step 3: Use exponential bounds

- Apply **Chernoff** for strong guarantees.

5.3 Key Insight

- These inequalities allow:
  - Bounding rare events
  - Designing systems with guarantees
  - Understanding worst-case scenarios

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6. Final Summary (Exam Revision)

Inequality	Uses	Strength
Markov	Expectation	Weak
Chebyshev	Mean + Variance	Moderate
Chernoff	Exponential moments	Strong
Jensen	Convexity	Structural

Final Takeaway

- All inequalities aim to bound:

$$P(\text{large deviation})$$

- Increasing strength:

$$\text{Markov} \to \text{Chebyshev} \to \text{Chernoff}$$

- Jensen provides a **fundamental tool** for expectation transformations.

If you want next, I can convert this **exact scribe into LaTeX code (verbatim, exam-ready)** like your previous requests.

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Here is the **precise, exam-oriented descriptive lecture scribe** starting **from Slide 18 (Markov's Inequality)** and continuing through all subsequent topics, strictly following the structure and logical progression of the provided slides .

Lecture Scribe: Tail Bounds (From Markov’s Inequality Onward)

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1. Markov’s Inequality

1.1 Definition

Let X be a **non-negative random variable** ($X \geq 0$).

For any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

1.2 Step-by-Step Proof

We derive the inequality directly from the definition of expectation.

Step 1: Start from expectation

$$\mathbb{E}[X] = \sum_x x \cdot P(X = x)$$

Step 2: Partition the sum

Split values into:

- $x < a$
- $x \geq a$

$$\mathbb{E}[X] = \sum_{x<a} xP(X = x) + \sum_{x \geq a} xP(X = x)$$

Step 3: Lower bound the second term

For all $x \geq a$,

$$x \geq a \Rightarrow xP(X = x) \geq aP(X = x)$$

Thus,

$$\sum_{x \geq a} xP(X = x) \geq a \sum_{x \geq a} P(X = x)$$

$$= a \cdot P(X \geq a)$$

Step 4: Combine with expectation

$$\mathbb{E}[X] \geq a \cdot P(X \geq a)$$

Step 5: Rearranging

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

1.3 Example

Let:

$$\mathbb{E}[X] = 10$$

We want:

$$P(X \geq 50)$$

Applying Markov:

$$P(X \geq 50) \leq \frac{10}{50} = 0.2$$

1.4 Interpretation

- Even without knowing distribution,
- We can **bound extreme values using expectation alone**.
- The bound is **general but loose**.

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2. Chebyshev's Inequality

2.1 Motivation

- Markov uses only expectation.
- To improve bounds, we incorporate **variance**.

2.2 Definition

Let X have:

- Mean $\mu = \mathbb{E}[X]$
- Variance $\sigma^2 = \mathbb{E}[(X - \mu)^2]$

Then for any $a > 0$,

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

2.3 Step-by-Step Proof (via Markov)

Step 1: Define a new random variable

$$Y = (X - \mu)^2$$

- $Y \geq 0$, so Markov applies.

Step 2: Apply Markov's inequality

$$P(Y \geq a^2) \leq \frac{\mathbb{E}[Y]}{a^2}$$

Step 3: Rewrite probability

$$P((X - \mu)^2 \geq a^2) = P(|X - \mu| \geq a)$$

Step 4: Substitute expectation

$$\mathbb{E}[Y] = \sigma^2$$

Final Result

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

2.4 Example

Let:

- $\mu = 100$
- $\sigma^2 = 25$

Then:

$$P(|X - 100| \geq 10) \leq \frac{25}{100} = 0.25$$

2.5 Interpretation

- Uses both **average and spread**.
- Provides tighter bounds than Markov.

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3. Chernoff Bound (Poisson Tail)

3.1 Motivation

- Markov and Chebyshev give polynomial bounds.
- Need **exponentially decreasing bounds** for rare events.

3.2 Core Idea

Apply Markov's inequality to:

$$e^{tX}$$

instead of X.

3.3 Derivation

Step 1: Transform the event

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

Step 3: Final bound

$$P(X \geq a) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

3.4 Interpretation

- $\mathbb{E}[e^{tX}]$ is the **moment generating function**.
- By choosing t, we get the **tightest bound**.

3.5 Insight

- Converts polynomial decay into **exponential decay**.
- Much stronger for bounding tail probabilities.

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4. Jensen's Inequality

4.1 Convex Functions

A function f is convex if:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

4.2 Definition

For a convex function f,

$$f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$$

4.3 Step-by-Step Proof

Step 1: Express expectation

$$\mathbb{E}[X] = \sum p_i x_i$$

Step 2: Apply convexity

$$f\left(\sum p_i x_i\right) \le \sum p_i f(x_i)$$

Step 3: Recognize expectation

$$\sum p_i f(x_i) = \mathbb{E}[f(X)]$$

Final Result

$$f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$$

4.4 Example Insight

For $f(x) = x^2$:

$$(\mathbb{E}[X])^2 \le \mathbb{E}[X^2]$$

4.5 Interpretation

- Expectation “smooths” values before applying convex function.
- Leads to inequality.

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5. Case Study: Worst-Case Server Demand

5.1 Problem

- Predict **maximum demand** on a server.
- Exact distribution unknown.

5.2 Approach

- Use inequalities to bound extreme events:

Step 1: Markov

- Uses expectation → rough bound

Step 2: Chebyshev

- Uses variance → tighter bound

Step 3: Chernoff

- Uses exponential → strongest bound

5.3 Insight

- Enables **system-level guarantees**.
- Helps in **capacity planning and risk estimation**.

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6. Final Exam Summary

Hierarchy of Bounds

$$\text{Markov} \;\rightarrow\; \text{Chebyshev} \;\rightarrow\; \text{Chernoff}$$

Key Formulas

- Markov:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

- Chebyshev:

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

- Chernoff:

$$P(X \geq a) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}$$

- Jensen:

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

Final Takeaway

- These inequalities allow bounding **tail probabilities without full distributions**.
- Increasing strength comes with **more information used**:
  - Expectation → Variance → Exponential moments

If you want next, I can generate the **exact LaTeX code version (verbatim)** for this scribe, matching your exam submission format.